

$$\begin{aligned}
 s+t &= (a_1v_1 + b_1v_2) + (a_2v_1 + b_2v_2) \\
 &= (a_1v_1 + b_1v_2 + a_2v_1) + b_2v_2 \text{ (Algebraic property (ii))} \\
 &= (a_1v_1 + (b_1v_2 + a_2v_1)) + b_2v_2 \text{ (Algebraic property (ii))} \\
 &= (a_1v_1 + (a_2v_1 + b_1v_2)) + b_2v_2 \text{ (Algebraic property (i))} \\
 &= ((a_1v_1 + a_2v_1) + b_1v_2) + b_2v_2 \text{ (Algebraic property (ii))} \\
 &= (a_1v_1 + a_2v_1) + (b_1v_2 + b_2v_2) \text{ (Algebraic property (ii))} \\
 &= (a_1 + a_2)v_1 + (b_1 + b_2)v_2 \text{ (Algebraic property (vi))}
 \end{aligned}$$

It's easy to see that $a_1 + a_2 \in \mathbb{R}$ and $b_1 + b_2 \in \mathbb{R}$.

$\therefore s+t$ is a linear combination of v_1 and v_2 . By defn, $s+t \in H$.

Let, $c \in \mathbb{R}$ and Let, $x \in H$. By Defn, x is a linear combination of v_1 and v_2 .

$$\begin{aligned}
 \therefore \exists s, t \in \mathbb{R} \text{ s.t. } x &= sv_1 + tv_2. \quad cx = c(sv_1 + tv_2) = c \cdot (sv_1) + c \cdot (tv_2) \\
 &\quad \text{(Algebraic property v)} \\
 &= (cs)v_1 + (ct)v_2 \text{ (Algebraic property vii)}
 \end{aligned}$$

It's easy to see that cs and $ct \in \mathbb{R}$. $\therefore cx$ is a linear combination of v_1 and v_2 . By Defn, $cx \in H$.

$\therefore$ From Defn, H is a subspace of $\mathbb{R}^n$. $\square$

If v_1 is not zero and if v_2 is a multiple of v_1 , then v_1 and v_2 simply span a line through the origin. It's easy to show that this is a subspace.

Example 2: A Line L not through the origin is not a subspace, because it does not contain the origin as required. Also, Figure 2 shows that L is not closed under addition or scalar multiplication. $\square$

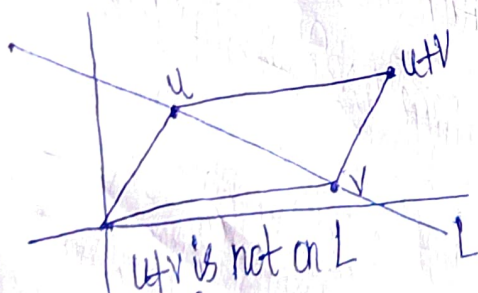
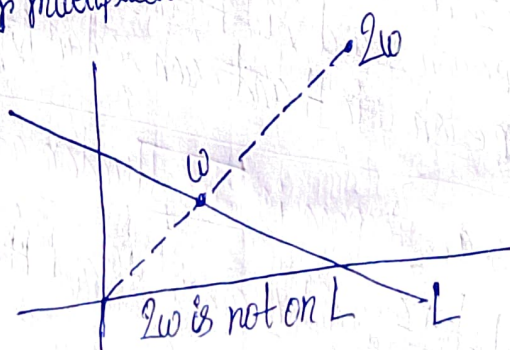


FIGURE 1



Example 3: For $v_1, \dots, v_p$ in $\mathbb{R}^n$, the set of all linear combinations of $v_1, \dots, v_p$ is a subspace of $\mathbb{R}^n$. The verification of this statement is similar to the argument given in Example 1. We shall now refer to $\text{Span}\{v_1, \dots, v_p\}$.